

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 1: BINARY SHAPE ANALYSIS

SLO2: Apply shape descriptors and morphological techniques to compare and analyze binary shapes for pattern recognition tasks.

ACTIVITY 1: SHAPE DESCRIPTOR ANALYSIS SOLUTIONS

1. Area of Shape A (number of object pixels) = **6**
2. Area of Shape B = **6**
3. Width of Shape A = **3**
4. Height of Shape B = **3**
5. Which shape has a larger aspect ratio? **Shape A** (Aspect Ratio of Shape A = $3/2 = 1.5$, Shape B = $2/3 = 0.67$)
6. Based on the shape descriptors, explain how these two shapes are different:

While both shapes have equal Area (6 pixels), Shape A is horizontally oriented with an aspect ratio of 1.5 (Width 3, Height 2), whereas Shape B is vertically oriented with an aspect ratio of 0.67 (Width 2, Height 3).

ACTIVITY 2: MORPHOLOGICAL OPERATIONS SOLUTIONS

1. If dilation is applied, what will happen to the object?

☒ **Object will expand**

Explanation: Dilation adds foreground pixels to the boundaries of binary objects, filling small holes and increasing the overall shape area.

2. If erosion is applied, what will happen to the object?

☒ **Object will shrink**

Explanation: Erosion removes pixels from object boundaries, stripping away boundary noise and reducing the overall shape size.

3. Why are morphological operations important before pattern recognition?

Morphological operations pre-process binary images by removing background noise, bridging small gaps, smoothing boundaries, and separating touching structures, which improves feature extraction accuracy.

REFLECTION QUESTION SOLUTIONS

How do shape descriptors and morphological operations together help in object classification?

Morphological operations clean and refine binary image structures (removing noise and connecting components) so that shape descriptors (such as area, aspect ratio, perimeter, and centroid) can be accurately extracted and measured to classify and identify distinct pattern classes effectively.